

contents

ELECTRONICS FOR YOU PLUS | JUNE 2015 | VOL. 4 NO. 02

TECH FOCUS

26 Turning the Hunky Into a Smartphone



BUYERS' GUIDE

68 Selecting the Right Wireless Module



18 Security

Electronics Surveillance: Successes and Failures

32 Tech Focus

Hybridisation and Thin-Film Sensors Key Enablers of Flexible Electronics Today

Embedded

40 Golden Rules for Designing Wearables

50 Electronic Products: In-the-Loop Testing Adds Embedded System Validation

54 What's New in Audio and Video Components

59 Infrastructure

Smartcities in India: A Long Way To Go

62 Test & Measurement

Touchscreen and Multi-Function Make Today's Oscilloscopes Powerful

72 Innovation

Raspberry Pi and M2M Technology Power This Smart Street Lighting System

74 EFY Plus DVD

This Month's DVD Contents

76 Interview

"I got my first eyeborg, which allowed me to sense colours" — Neil Harbisson, government-recognised cyborg

DO-IT-YOURSELF



- Solar Emergency Lamp 80
- GPS Master-Slave Clocks With RF Link 83
- Radio Frequency Detector 88
- AC 230V Electromagnetic Relay 89
- Multichannel Visual Tachometer 91
- Glass Break Alarm 92
- Three Amazing Things You Can Do With Raspberry Pi 94
- Sixth-Sense Media Player 98
- Kansas Lava to Simulate Circuits 102

EFY Plus DVD

- CAELinux: A Distribution of Open Source Engineering Tools I
- Audio Analysis Made Easy Using Essential IV
- Smart Deeds for Mastering Digital Electronics VI

78 Interview

"We put in a 15-minute time delay... to take care of last-minute order changes" — Glen Smith, CEO, Mouser Electronics

eStyle

108 Ten Android Apps for Everyday Use on Your Smartphone

110 Buyers' Guide: The Right Wearable for Your Wrist

117 Make in India

Market Survey: Training Sector Seeing Rising Demand for New Subjects

EDITOR	: Ramesh Chopra
EDITORIAL CORRESPONDENCE	: Editorial Secretary Phone: 011-26810601; E-mail: editsec@efy.in (Technical queries: eflyab@efy.in)
SUBSCRIPTIONS & MISSING ISSUES	: Phone: 011-26810601 or 02 or 03 E-mail: support@efy.in
BACK ISSUES, BOOKS, CDs, PCBs etc.	: Kit's Spares, New Delhi Phone: 011-26371661, 26371662 E-mail: info@kitspares.com
EXCLUSIVE NEWS/STANDOUT DISTRIBUTOR	: IBH Books & Magazines Distributors Ltd Unit No. 10, Bezzola Complex, Opp. Suman Nagar, Son Trombay Road, Chembur, Mumbai 400 071 Phone: 022-4049740/02
ADVERTISEMENTS	
NEW DELHI (HEAD OFFICE)	: Ph: 011-26810601 or 02 or 03 E-mail: eflyab@efy.in
MUMBAI	: Ph: 022-24950047, 24928520 E-mail: elyumn@efy.in
BENGALURU	: Ph: 080-25363394, 25260023 E-mail: eflybng@efy.in
CHENNAI	: Ph: 044-42994363 E-mail: eflychennai@efy.in
HYDERABAD	: Ph: 040-67172633 E-mail: eflyhyderabad@efy.in
KOLKATA	: Ph: 033-22294788 E-mail: eflykolkata@efy.in
PUNE	: Ph: 020-40147882 E-mail: elypune@efy.in
GUJARAT	: Ph: 079-61344948 E-mail: eflyahd@efy.in
CHINA	: Power Pioneer Group Inc. Ph: (86-755) 83727977, (86) 1392802595 E-mail: powerpioneer@efy.in
JAPAN	: Tandem Inc., Ph: 81-3-3541-4166 E-mail: tandem@efy.in
SINGAPORE	: Publicitas Singapore Pte Ltd Ph: +65-6836-2272 E-mail: publicitas@efy.in
TAIWAN	: J.K. Media, Ph: 886-2-8726780 ext. 10 E-mail: jkmedia@efy.in
UNITED STATES	: E & Tech Media Ph: +1 800 336 6677 E-mail: venoniquelamarque@gmail.com

SUBSCRIPTION RATES					
Period	Newstand Price (₹)	You Pay (₹)	Digital (₹)	Overseas Digital	Print
Two	2400	1800	750	US\$ 15	—
One	1200	960	280	US\$ 5	US\$ 120

Please send payments only in favour of **EFY Enterprises Pvt Ltd**
Valid till August 31, 2015, only

Printed, published and owned by Ramesh Chopra. Printed at Nutech Photooffsets, B-18, Okhla Industrial Area, Phase-1, New Delhi, on the first day of each month and published from D-37/1, Okhla Industrial Area, Phase-1, New Delhi 11020. Copyright 2015. All rights reserved throughout the world. No reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any errors due to publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrong claims made by advertisers. Disputes, if any, will be settled in a New Delhi court only.

Regulars

- 06 Feedback
- 08 Q&A
- 10 Useful Websites
- 12 Tech News
- 106 First Look
- 113 Make in India: Industry News
- 126 New Products
- 129 Business Pages Ads
- 139 Electronics Mart Ads
- 144 Product Categories Index + Attractions During 2015
- 145 Advertisers' Index

NEXT ISSUE • Smartcars • Automotive Electronics • Budget-Friendly Oscilloscopes